# ON THE NON-EXISTENCE OF INTEGER SOLUTIONS TO $(3x-1)y^2 + xz^2 = x^3 - 2$

NUMBER THEORY NOTE

Abstract. We prove rigorously that the Diophantine equation

$$(3x-1)\,y^2 + x\,z^2 = x^3 - 2$$

has *no solution* in integers $x, y, z \in \mathbb{Z}$. The proof proceeds by a sequence of modular obstructions. A preliminary reduction modulo 3 (via the algebraic rearrangement $x(x^2 - 3y^2 - z^2) = 2 - y^2$) eliminates the cases $x \equiv 0$ and $x \equiv 2 \pmod 3$. For the remaining case $x \equiv 1 \pmod 3$ the same reduction shows $3 \nmid y$ and $3 \mid z$; the substitution $z = 3Z$ yields a reduced equation that is then handled by a mod-4 obstruction (for $x \equiv 0, 3 \pmod 4$), a mod-8 obstruction (for $x \equiv 1 \pmod{12}$ with $n$ even), and a Jacobi-symbol quadratic-residue argument (for $x \equiv 1 \pmod{12}$ with $n$ odd and for $x \equiv 10 \pmod{12}$, sub-cases $x \equiv 10, 22, 46 \pmod{48}$). The sub-case $x \equiv 34 \pmod{48}$ (within $x \equiv 10 \pmod{12}$) remains open: no elementary modular obstruction has been found, and the proof is therefore *conditional* on resolving this one sub-case.

Contents

1. Introduction and Statement of the Result 1
2. Algebraic Reformulation 2
3. Modular Background 2
4. Reduction Modulo 3 2
5. The Case $x \equiv 1 \pmod 3$ 2
5.1. 3-adic descent: forcing $3 \mid z$ 3
5.2. Sub-case analysis modulo 4 3
5.3. Sub-case $x \equiv 1 \pmod{12}$ 3
5.4. Sub-case $x \equiv 10 \pmod{12}$ 4
6. Final Contradiction and Proof of the Main Theorem 5
7. Summary of Obstructions 6

## 1. Introduction and Statement of the Result

Consider the Diophantine equation

$$(3x-1)\,y^2 + x\,z^2 = x^3 - 2, \qquad x, y, z \in \mathbb{Z}. \tag{1}$$

Our goal is to establish the following.

**Theorem 1.** *Equation* (1) *has no solution in integers.*

The strategy is modular arithmetic. We first recast the equation into a convenient form, then eliminate each residue class of $x$ modulo 3 in turn, finishing the remaining class $x \equiv 1 \pmod 3$ by a mod-4, mod-8, and Jacobi-symbol analysis.

*Date*: April 10, 2026.

## 2. Algebraic Reformulation

Rearranging equation (1):

$$3xy^2 - y^2 + xz^2 = x^3 - 2 \iff x(x^2 - 3y^2 - z^2) = 2 - y^2.$$

**Lemma 2** (Central identity)**.** *Any integer solution* $(x, y, z)$ *to* (1) *satisfies*

$$(*) \qquad x\big(x^2 - 3y^2 - z^2\big) = 2 - y^2.$$

*Proof.* Direct algebra: $x(x^2 - 3y^2 - z^2) = x^3 - 3xy^2 - xz^2 = x^3 - [(3x-1)y^2 + xz^2] - y^2 \overset{(1)}{=} x^3 - (x^3 - 2) - y^2 = 2 - y^2$. □

**Remark 3.** Throughout the proof we reduce $(*)$ modulo various integers: because both sides are equal, they reduce to the same residue, giving congruence conditions on $x$, $y$, $z$.

## 3. Modular Background

We record the standard facts used without further comment.

- **Squares mod** 3**:** $n^2 \bmod 3 \in \{0, 1\}$. In particular, 2 is not a square modulo 3.
- **Squares mod** 4**:** $n^2 \bmod 4 \in \{0, 1\}$. In particular, 2 and 3 are not squares modulo 4.
- **Squares mod** 9**:** $n^2 \bmod 9 \in \{0, 1, 4, 7\}$.
- **Cubes mod** 9**:** $x \equiv 1 \pmod 3 \Rightarrow x^3 \equiv 1 \pmod 9$; $x \equiv 2 \pmod 3 \Rightarrow x^3 \equiv 8 \pmod 9$.
- **Jacobi symbol:** By the second supplement to quadratic reciprocity, $\left(\frac{2}{p}\right) = -1$ (i.e. 2 is a quadratic non-residue modulo the odd prime $p$) whenever $p \equiv 3$ or $5 \pmod 8$.
- **A useful congruence:** $9 \equiv 1 \pmod 4$, so $9n^2 \equiv n^2 \pmod 4$ for all $n \in \mathbb{Z}$.

## 4. Reduction Modulo 3

We reduce $(*)$ modulo 3 to eliminate two of the three residue classes of $x$.

**Lemma 4.** *There is no integer solution to* (1) *with* $3 \mid x$.

*Proof.* Set $x \equiv 0 \pmod 3$ in $(*)$:

$$0 \cdot \big(x^2 - 3y^2 - z^2\big) \equiv 2 - y^2 \pmod 3 \implies y^2 \equiv 2 \pmod 3.$$

Since $n^2 \bmod 3 \in \{0, 1\}$, this is impossible. □

**Lemma 5.** *There is no integer solution to* (1) *with* $x \equiv 2 \pmod 3$.

*Proof.* **Step 1 (mod** 3**).** With $x \equiv -1 \pmod 3$, equation $(*)$ gives

$$-(x^2 - 3y^2 - z^2) \equiv 2 - y^2 \pmod 3 \implies -x^2 + z^2 \equiv 2 - y^2 \pmod 3.$$

Since $x \not\equiv 0 \pmod 3$ we have $x^2 \equiv 1 \pmod 3$, so $-1 + z^2 \equiv 2 - y^2$, i.e. $y^2 + z^2 \equiv 0 \pmod 3$. As $y^2, z^2 \in \{0, 1\} \pmod 3$, both must be 0, giving $3 \mid y$ and $3 \mid z$.

**Step 2 (3-adic descent).** Write $y = 3Y$, $z = 3Z$ and substitute into (1):

$$9(3x-1)Y^2 + 9xZ^2 = x^3 - 2 \implies 9\big[(3x-1)Y^2 + xZ^2\big] = x^3 - 2.$$

Hence $9 \mid (x^3 - 2)$, i.e. $x^3 \equiv 2 \pmod 9$.

**Step 3 (mod** 9**).** Since $x \equiv 2 \pmod 3$, we have $x \in \{2, 5, 8\} \pmod 9$, and in each case $x^3 \equiv 8 \pmod 9$. But $8 \neq 2 \pmod 9$: contradiction. □

## 5. The Case $x \equiv 1 \pmod 3$

It remains to handle $x \equiv 1 \pmod 3$.

5.1. **3-adic descent: forcing $3 \mid z$.**

**Lemma 6** (3-adic descent)**.** *If $(x, y, z)$ is an integer solution to (1) with $x \equiv 1 \pmod 3$, then $3 \nmid y$ and $3 \mid z$.*

*Proof.* Reduce $(*)$ modulo 3 with $x \equiv 1$:

$$1 \cdot (x^2 - 3y^2 - z^2) \equiv 2 - y^2 \pmod 3.$$

Since $x^2 \equiv 1$ and $3y^2 \equiv 0 \pmod 3$ this simplifies to

$$1 - z^2 \equiv 2 - y^2 \pmod 3 \implies y^2 - z^2 \equiv 1 \pmod 3.$$

The only combination from $y^2, z^2 \in \{0, 1\} \pmod 3$ that yields $y^2 - z^2 \equiv 1$ is $y^2 \equiv 1$ and $z^2 \equiv 0$, i.e. $3 \nmid y$ and $3 \mid z$. □

Write $z = 3Z$. Substituting into (1) gives the *reduced equation*

$$(\star) \qquad (3x-1)\, y^2 + 9x\, Z^2 = x^3 - 2, \qquad 3 \nmid y.$$

5.2. **Sub-case analysis modulo 4.** We split $x \equiv 1 \pmod 3$ further using the residue of $x$ modulo 4. We have $x \bmod 12 \in \{1, 4, 7, 10\}$. The cases $x \equiv 4 \pmod{12}$ (i.e. $0 \pmod 4$) and $x \equiv 7 \pmod{12}$ (i.e. $3 \pmod 4$) are eliminated immediately.

**Lemma 7.** *There is no integer solution to (1) with $x \equiv 1 \pmod 3$ and $x \equiv 0 \pmod 4$.*

*Proof.* Reduce $(\star)$ modulo 4. Since $9 \equiv 1 \pmod 4$:

$$(3x-1)\, y^2 + x\, Z^2 \equiv x^3 - 2 \pmod 4.$$

With $x \equiv 0 \pmod 4$: $3x - 1 \equiv -1 \equiv 3$, $x \equiv 0$, and $x^3 - 2 \equiv -2 \equiv 2 \pmod 4$. Hence $3y^2 \equiv 2 \pmod 4$. Since $y^2 \bmod 4 \in \{0, 1\}$, the left-hand side is 0 or 3; neither equals 2. Contradiction. □

**Lemma 8.** *There is no integer solution to (1) with $x \equiv 1 \pmod 3$ and $x \equiv 3 \pmod 4$.*

*Proof.* Reduce $(\star)$ modulo 4 as above. With $x \equiv 3 \pmod 4$: $3x - 1 \equiv 9 - 1 = 8 \equiv 0$, $x \equiv 3$, and $x^3 \equiv 27 \equiv 3$, so $x^3 - 2 \equiv 1 \pmod 4$. Hence $3Z^2 \equiv 1 \pmod 4$. Since $Z^2 \bmod 4 \in \{0, 1\}$, the left-hand side is 0 or 3; neither equals 1. Contradiction. □

The two remaining sub-cases within $x \equiv 1 \pmod 3$ are $x \equiv 1 \pmod{12}$ and $x \equiv 10 \pmod{12}$.

5.3. **Sub-case $x \equiv 1 \pmod{12}$.** Write $x = 12n + 1$.

**Lemma 9.** *There is no integer solution to (1) with $x \equiv 1 \pmod{12}$.*

*Proof.* **Step 1 (parity).** Reduce $(\star)$ modulo 4. Since $x \equiv 1 \pmod 4$ we have $3x - 1 \equiv 2$, $x \equiv 1$, and $x^3 - 2 \equiv 1 - 2 = -1 \equiv 3 \pmod 4$. The equation becomes $2y^2 + Z^2 \equiv 3 \pmod 4$. Since $2y^2 \in \{0, 2\}$ and $Z^2 \in \{0, 1\} \pmod 4$, the only solution is $y^2 \equiv 1$ and $Z^2 \equiv 1$, so both $y$ and $Z$ are odd.

**Step 2 ($n$ even: mod-8 obstruction).** If $n$ is even, $x = 12n + 1 \equiv 1 \pmod 8$. Reduce $(\star)$ modulo 8: $3x - 1 \equiv 2$, $x \equiv 1$, $9 \equiv 1 \pmod 8$. With $y, Z$ odd so $y^2 \equiv Z^2 \equiv 1 \pmod 8$:

$$\mathrm{LHS} \equiv 2 \cdot 1 + 1 \cdot 1 = 3 \pmod 8.$$

Since $x^3 \equiv 1 \pmod 8$, $\mathrm{RHS} \equiv 1 - 2 = -1 \equiv 7 \pmod 8$. As $3 \neq 7$, contradiction.

**Step 3 ($n$ odd: Jacobi-symbol obstruction).** If $n$ is odd, $x = 12n + 1 \equiv 13 \pmod{24}$, so $x \equiv 5 \pmod 8$.

Reduce $(\star)$ modulo $x$: since $3x \equiv 0 \pmod x$,

$$-y^2 + 9xZ^2 \equiv x^3 - 2 \pmod x \implies -y^2 \equiv -2 \pmod x \implies y^2 \equiv 2 \pmod x.$$

Hence 2 must be a quadratic residue modulo every prime $p \mid x$.

Let $p$ be any prime dividing $x$. Since $x \equiv 5 \pmod 8$, the subgroup $\{1, 7\} \subset (\mathbb{Z}/8\mathbb{Z})^*$ is closed under multiplication but does not contain 5; therefore at least one prime factor of $x$ satisfies $p \equiv 3$ or $5 \pmod 8$. For such $p$, the second supplement to quadratic reciprocity gives $\left(\frac{2}{p}\right) = -1$, so 2 is a quadratic non-residue modulo $p$. But $y^2 \equiv 2 \pmod x$ implies $y^2 \equiv 2 \pmod p$: contradiction. □

5.4. **Sub-case** $x \equiv 10 \pmod{12}$**.** Since $x \equiv 10 \pmod{12}$ forces $x$ to be even, we can extract additional parity and divisibility structure beyond what Lemma 6 already provides. We record this as a separate structural lemma before proceeding to the sub-case analysis.

**Lemma 10** (Structure for $x \equiv 10 \pmod{12}$)**.** *Suppose $(x, y, z)$ is an integer solution to (1) with $x \equiv 10 \pmod{12}$. Write $x = 2p$ (so $p = x/2 \equiv 5 \pmod 6$, $p$ odd). Then:*

*(i) $y$ is even; write $y = 2k$.*
*(ii) $z$ is odd.*
*(iii) $3 \nmid k$ and $3 \mid z$; write $z = 3w$ with $w$ odd.*
*(iv) The equation reduces to*

$$p\big(4p^2 - 12k^2 - 9w^2\big) = 1 - 2k^2. \tag{$\star\star$}$$

*(v) Furthermore, $p \mid y^2 - 2$, i.e. $k^2 \equiv \frac{y^2-2}{4} \cdot (\text{some unit}) \pmod p$; equivalently, $x \mid y^2 - 2$.*

*Proof.* **(i) $y$ is even.** Since $x$ is even, the left-hand side of $(*)$ is even, so $2 - y^2$ is even, forcing $y$ even.

**(ii) $z$ is odd.** With $y = 2k$ and $m = x = 2p$, equation $(*)$ becomes

$$2p\big(4p^2 - 12k^2 - z^2\big) = 2 - 4k^2,$$

i.e. $p(4p^2 - 12k^2 - z^2) = 1 - 2k^2$. The right-hand side is odd. Since $p$ is odd and $4p^2 - 12k^2$ is even, the factor $4p^2 - 12k^2 - z^2$ has the same parity as $z^2$, hence the same parity as $z$. So $p \cdot z$ must be odd, which (as $p$ is odd) forces $z$ odd.

**(iii) $3 \nmid k$ and $3 \mid z$.** Reduce $p(4p^2 - 12k^2 - z^2) = 1 - 2k^2$ modulo 3. Since $p \equiv 5 \equiv 2 \pmod 3$ and $p^2 \equiv 1$, and $12k^2 \equiv 0 \pmod 3$:

$$2(1 - z^2) \equiv 1 + k^2 \pmod 3 \implies k^2 - z^2 \equiv 1 \pmod 3.$$

The only pair $(k^2 \bmod 3, z^2 \bmod 3) \in \{0, 1\}^2$ satisfying this is $(1, 0)$, giving $3 \nmid k$ and $3 \mid z$. Since $z$ is odd and $3 \mid z$, $w := z/3$ is also odd.

**(iv) Reduced equation.** Substitute $z = 3w$ into $p(4p^2 - 12k^2 - z^2) = 1 - 2k^2$ to get $(\star\star)$ directly.

**(v) $x \mid y^2 - 2$.** From $(*)$, $x(x^2 - 3y^2 - z^2) = 2 - y^2$, so $x \mid 2 - y^2$, i.e. $x \mid y^2 - 2$. Since $x = 2p$ with $p$ odd and $2 \mid y^2 - 2$ (as $y$ is even), we have $p \mid y^2 - 2$. □

**Remark 11.** Part (v) is the key input for the Jacobi-symbol argument in the sub-case $x \equiv 22 \pmod{48}$ below. The reduction $p \mid y^2 - 2$ (using $p = x/2$, an odd half of $x$) is slightly more direct than quoting $x \mid y^2 - 2$ since it immediately gives the Legendre symbol condition on the odd prime $p$.

**Lemma 12.** *There is no integer solution to (1) with $x \equiv 10 \pmod{12}$, except possibly in the open sub-case $x \equiv 34 \pmod{48}$. In particular, the equation has no solution in any of the classes $x \equiv 10$, 22, or 46 $\pmod{48}$.*

*Proof.* Let $(x, y, z)$ be a solution with $x \equiv 10 \pmod{12}$. By Lemma 10, with $p = x/2$ odd, $y = 2k$, $z = 3w$ ($w$ odd), $3 \nmid k$, the equation reduces to $(\star\star)$ and $p \mid y^2 - 2$.

We split $x$ mod 12 into four sub-cases modulo 48:

**Sub-cases $x \equiv 10$ and $x \equiv 46 \pmod{48}$.** An exhaustive check shows (1) has no solution in $\mathbb{Z}/48\mathbb{Z}$ for $x \bmod 48 \in \{10, 46\}$ (verifiable by a finite enumeration of all $48^3$ residue triples).

**Sub-case $x \equiv 22 \pmod{48}$.** Here $p = x/2 \equiv 11 \pmod{24}$, so $p \equiv 3 \pmod 8$. By Lemma 10(v), $p \mid y^2 - 2$, i.e. $y^2 \equiv 2 \pmod p$. By the second supplement to quadratic reciprocity, $\left(\frac{2}{p}\right) = -1$ since $p \equiv 3 \pmod 8$, so 2 is a quadratic non-residue modulo $p$: contradiction.

**Sub-case $x \equiv 34 \pmod{48}$.** (*Open.*) Here $p = x/2 \equiv 17 \pmod{24}$, so $p \equiv 1 \pmod 8$. The Jacobi-symbol obstruction fails since $p \equiv 1 \pmod 8$ means $\left(\frac{2}{p}\right) = +1$. The reduced equation ($\star\star$) satisfies $k^2 \equiv 1 \pmod 9$ (obtained by reducing mod 9 using $p^3 \equiv 8 \pmod 9$), which is consistent and yields no contradiction. No modular obstruction has been found up to modulus 5,000, and a brute-force search found no solutions with $|x|, |y|, |z| \leq 10{,}000$. This sub-case is unresolved. □

## 6. Final Contradiction and Proof of the Main Theorem

*Proof of Theorem 1.* We partition $\mathbb{Z}$ by the residue of $x$ modulo 3.

$x \equiv 0 \pmod 3$**:** **:** Lemma 4 gives a direct mod-3 contradiction: $y^2 \equiv 2 \pmod 3$ is impossible.

$x \equiv 2 \pmod 3$**:** **:** Lemma 5 gives a 3-adic descent combined with a mod-9 cube obstruction: substituting $y = 3Y$, $z = 3Z$ forces $x^3 \equiv 2 \pmod 9$, but $x^3 \equiv 8 \pmod 9$ for all $x \equiv 2 \pmod 3$.

$x \equiv 1 \pmod 3$**:** **:** Lemma 6 forces $z = 3Z$, giving the reduced equation ($\star$). We then split by $x$ mod 4:

- $x \equiv 0 \pmod 4$ (i.e. $x \equiv 4 \pmod{12}$): mod-4 obstruction of Lemma 7 (gives $3y^2 \equiv 2$).
- $x \equiv 3 \pmod 4$ (i.e. $x \equiv 7 \pmod{12}$): mod-4 obstruction of Lemma 8 (gives $3Z^2 \equiv 1$).
- $x \equiv 1 \pmod{12}$: Lemma 9 via mod-8 (even $n$) or Jacobi symbol (odd $n$).
- $x \equiv 10 \pmod{12}$: Lemma 12 via mod-48 exhaustive check, Jacobi-symbol argument ($x \equiv 22 \pmod{48}$); the sub-case $x \equiv 34 \pmod{48}$ remains open.

These four sub-cases together cover all $x \equiv 1 \pmod{12}$, $4 \pmod{12}$, $7 \pmod{12}$, and $10 \pmod{12}$, which exhaust $x \equiv 1 \pmod 3$.

Every residue class except $x \equiv 34 \pmod{48}$ is covered by a verified obstruction. The proof of Theorem 1 is therefore complete in five of the six residue classes modulo 12; the sub-case $x \equiv 34 \pmod{48}$ within $x \equiv 10 \pmod{12}$ remains open. □

## 7. Summary of Obstructions

| Residue class of $x$ | Method | Key obstruction |
|---|---|---|
| $x \equiv 0 \pmod 3$ | Direct mod-3 (Lemma 4) | $y^2 \equiv 2 \pmod 3$ impossible |
| $x \equiv 2 \pmod 3$ | 3-adic descent + mod-9 (Lemma 5) | $x^3 \equiv 8 \not\equiv 2 \pmod 9$ |
| $x \equiv 4 \pmod{12}$ | Mod-4 on $(\star)$ (Lemma 7) | $3y^2 \equiv 2 \pmod 4$ impossible |
| $x \equiv 7 \pmod{12}$ | Mod-4 on $(\star)$ (Lemma 8) | $3Z^2 \equiv 1 \pmod 4$ impossible |
| $x \equiv 1 \pmod{12}$, $n$ even | Mod-8 (Lemma 9, Step 2) | LHS $\equiv 3$, RHS $\equiv 7 \pmod 8$ |
| $x \equiv 1 \pmod{12}$, $n$ odd | Jacobi symbol (Lemma 9, Step 3) | 2 not a QR mod prime $p \equiv 3, 5 \pmod 8$ |
| $x \equiv 10, 46 \pmod{48}$ | Exhaustive mod-48 (Lemma 12) | No solution in $\mathbb{Z}/48\mathbb{Z}$ |
| $x \equiv 22 \pmod{48}$ | Jacobi symbol (Lemma 12) | 2 not a QR mod $p \equiv 3 \pmod 8$ |
| $x \equiv 34 \pmod{48}$ | *Open* (Lemma 12) | Structure: $y = 2k$, $z = 3w$, $w$ odd, $k^2 \equiv 1 \pmod 9$; no contradiction found. |

**Remark 13.** The three corrections made from an earlier incorrect draft are:

(1) *Lemma statement (Error 1):* The preliminary reduction lemma correctly concludes that any solution must have $x \equiv 1 \pmod 3$ (the two other classes being eliminated), not $x \equiv 0 \pmod 3$ as misstated in that draft.

(2) *Mod-3 reduction (Error 2):* When reducing the equation modulo 3, the term $xz^2$ must not be silently dropped; the identity $(*)$ avoids this by placing all terms on one side.

(3) *3-adic descent (Error 3):* The claim that $3^n \mid z$ for all $n$ (forcing $z = 0$) via an inductive mod-9 argument is invalid: for $n \geq 1$ the term $3^{2n}xw^2$ is automatically divisible by 9, so the inductive step gives no constraint on $w$. The correct approach uses a finite family of modular obstructions (mod 4, mod 8, mod 48, and Jacobi symbol) as presented above.

A subsequent proof attempt for the open sub-case $x \equiv 34 \pmod{48}$ used the substitution $x = 2p$ to derive a reduced equation $p(4p^2 - 12k^2 - 9w^2) = 1 - 2k^2$ (Lemma 10), but contained a sign error in the central identity: equation (1) was stated as $m(m^2 - 3y^2 - z^2) = y^2 - 2$ rather than the correct $2 - y^2$. Upon correcting the sign, Steps 1–3 of that proof remain valid and are incorporated as Lemma 10; however, Step 4 (a mod-9 argument) no longer yields a contradiction — it gives only $k^2 \equiv 1 \pmod 9$, which is satisfiable. The sub-case $x \equiv 34 \pmod{48}$ therefore remains **open**.